# MILITARY SPECIFICATION

## ETHYLCELLULOSE, TECHNICAL (METRIC)

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers one technical grade of ethylcellulose.

1.2 Classification. Ethylcellulose shall be of the following classes as specified (see 6.2):

Class 1 - No. 10 viscosity
Class 2 - No. 7 viscosity

## 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieve, Test
PPP-D-729 - Drums, Shipping and Storage, Steel, 55-Gallon (208 Liters)

FSC 6810

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-B-117 - Bags, Sleeve and Tubing - Interior Packaging

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads for 40" by 48" Pallets
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D445 - Kinematic Viscosity of Transparent and Opaque Liquids (and the Calculation of Dynamic Viscosity)
D914 - Ethylcellulose
D1193 - Reagent Water

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. Ethylcellulose shall be white porous granules when tested as specified in 4.2.4.1.

3.2 Chemical and physical characteristics. Ethylcellulose shall conform to the applicable chemical and physical characteristics of table I when tested as specified therein.



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TABLE I. Chemical and physical characteristics

| Characteristic | Class 1 | Class 2 | Test paragraph |
|---|---|---|---|
| Moisture, maximum percent by weight | 2.2 | 2.2 | 4.2.4.2 |
| Ash, sulfated, maximum percent by weight (dry basis) | 0.35 | 0.35 | 4.2.4.3 |
| Chloride, maximum percent by weight as NaCl (dry basis) | — | 0.10 | 4.2.4.4 |
| Acidity | None | None | 4.2.4.5 |
| Ethoxyl content, percent by weight (dry basis) | 48.5 ± 1.0 | 48.5 ± 1.0 | 4.2.4.6 |
| Viscosity at 25°C of 5 percent by weight in a 4 to 1 toluene-ethanol solution, centipoises | 10 ± 1 | 7 ± 1 | 4.2.4.7 |
| Percent by weight material retained on a No. 10 sieve | — | 1.0 | 4.2.4.8 |

3.3 Thermoplastic point. Class 1 ethylcellulose shall fuse without charring and shall be capable of being drawn into a thread at a temperature no less than 145°C when tested as specified in 4.2.4.9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the ethylcellulose produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).



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4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For ethylcellulose test. Sampling shall be conducted in accordance with table II. A representative specimen of approximately 400 grams (g) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE II. Sampling for test

| Number of containers in batch or lot | Number of sample containers |
|---|---|
| 2 to 25 | 2 |
| 26 to 150 | 3 |
| 151 to 1,200 | 5 |
| 1,201 to 7,000 | 8 |
| 7,001 to 20,000 | 10 |
| Over 20,000 | 20 |

4.2.2.3 For container leakage test. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shll be one filled unit, intermediate, or shipping container, as applicable, ready for shipment. Sample unit, intermediate, and shipping containers shall be examined for the following defects grouped collectively using an AQL of 4.0 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Marking incorrect, missing, or illegible
(f) Liner bag missing or not as specified
(g) Unitization not as specified

4.2.3.2 For ethylcellulose test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.3.3 For container leakage test. The sample unit shall be one container. The sample containers selected in 4.2.2.3 shall be tested as specified in 4.2.5 using an AQL of 2.5 percent defective.



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4.2.4 Ethylcellulose tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Material. Visually examine the specimen for form and color.

4.2.4.2 Moisture. Determine the percent by weight moisture in the specimen in accordance with the test procedure for moisture in ASTM D914 except repeat the heating until constant weight is obtained.

4.2.4.3 Ash. Dry approximately 5 g of the specimen to constant weight at 100° to 105°C. Determine the percent by weight ash in the dried specimen as sulfate in accordance with the test procedure for sulfated ash in ASTM D914.

4.2.4.4 Chloride (class 2 only). Determine the parts per million chloride (ppm) in the specimen in accordance with the test procedure for chloride in ASTM D914. Convert to percent by weight by dividing the ppm by 10,000.

4.2.4.5 Acidity. Boil 10 g of the specimen with 150 milliliters (ml) of water for 5 minutes, allow to settle, and cool. Use methyl red indicator to determine the presence of any acidity.

4.2.4.6 Ethoxyl content. Determine the percent by weight ethoxyl content in the specimen in accordance with the test procedure for ethoxyl content in ASTM D914.

4.2.4.7 Viscosity. Determine the dynamic viscosity of a 5 percent by weight solution of the previously dried specimen in a 4 to 1 toluene-ethanol solvent system at 25°C in accordance with ASTM D445. The density of the solution may be determined by any method accurate to three significant figures.

4.2.4.8 Material retained on a No. 10 sieve (class 2 only). Weigh approximately 100 g of the specimen to the nearest 0.1 g and place on a No. 10 sieve conforming to RR-S-366 and equipped with a cover and a bottom pan. Shake the sieve for 5 minutes in a mechanical shaker geared to produce 300 ± 15 gyrations and 150 ± 10 taps of the striker per minute. Weigh the portion retained on the sieve and calculate the percent by weight retained.

4.2.4.9 Thermoplastic point (class 1 only). Fill a 20 by 150 mm test tube to a depth of 25 mm with the specimen, ground to pass a 16-mesh sieve. Submerge the tube to a depth of 50 mm in an oil bath previously heated to 120°C. Place a thermometer in the specimen and increase the bath temperature at the rate of 2°C per minute. Stir the specimen with the thermometer during the heating. At a temperature approximately 25°C below the fusion point the particles soften and tend to cohere. The specimen becomes readily compressible and can no longer be stirred freely. Continue heating and working the mass until it fuses and becomes plastic. Read the thermoplastic point at that temperature at which the specimen can be repeatedly drawn into a thread by



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pulling the thermometer from the plastic mass. Duplicate determinations should check within 3°C.

4.2.3 Container leakage test. Place the sample drum on its side, roll it a distance of approximately 2 meters forwards and 2 meters backwards. Open the drum and inspect the bag liner for evidence of leakage (sifting of contents from the bag).

5. PACKAGING

5.1 Unit packing. Unit packing shall be level A, level B, or commercial as specified (see 6.2).

5.1.1 Level A. A quantity of ethylcellulose as specified (see 6.2) shall be unit packed in a close-fitting, fully removable head, with bolted locking ring steel drum with bag liner. The drum shall, as a minimum, conform to the Uniform Freight Classification (UFC) Rule 40. In addition, the exterior of the drum shall be prepared and coated as specified for the drums of PPP-D-729. The bag liner shall be designed to fit the drum and shall conform to MIL-B-117, type I, class E, style optional. The bag shall be sealed as specified in MIL-B-117 and the drum shall be closed in accordance with the drum specification. There shall be no evidence of sifting of contents from the bag when tested as specified in 4.2.5.

5.1.2 Level B. Ethylcellulose shall be unit packed level B as specified for level A in 5.1.1 except that the drum shall be of fiber construction conforming, as a minimum, to UFC Rule 51.

5.1.3 Commerical. A quantity of ethylcellulose as specified (see 6.2) shall be unit packed commercially in accordance with MIL-STD-1188.

5.2 Packing. Ethylcellulose unit packed as specified in 5.1 shall require no further protection, except for unitization (when specified).

5.3 Unitization. When specified (see 6.2), level A and B packs of ethyl cellulose shall be palletized in accordance with MIL-STD-147. Commercial packs shall be palletized in a manner to assure carrier acceptance and safe delivery at destination.

5.4 Marking. Level A and B unit packs and pallet loads shall be marked in accordance with MIL-STD-129. Commercial and unit packs shall be marked in accordance with MIL-STD-1188. Commercial pallet loads shall be marked to assure proper identification and prompt arrival at destination. Containers shall be marked to show date of manufacture and lot or batch number.

6. NOTES

6.1 Intended use. The ethylcellulose covered by this specification is intended for use in the manufacture of certain munition components. Class 1



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material is intended for use as a thermoplastic molding or extrusion compound. Class 2 material is intended for use as an ingredient in the binder for propellant powder.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Class of ethyl cellulose required (see 1.2)
(c) Level of unit packing required (see 5.1)
(d) Unit quantity required (see 5.1.1 and 5.1.3)
(e) If palletization is required (see 5.3)

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

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